Fall 2024 - IST 686 - Final Examination Data Prep

Your name here

Data

You have a personalized RData file available on Blackboard that contains two data sets that pertain to vaccinations for the United States as a whole and for Californian school districts. The U.S. vaccine data is a time series and the California data is a sample of end-of-year vaccination reports from n=700 school districts. Here is a description of the datasets:

usVaccines – Time series data from the World Health Organization reporting vaccination rates in the U.S. for five common vaccines

```
Time-Series [1:38, 1:5] from 1980 to 2017:
 - attr(*, "dimnames")=List of 2
  ..$ : NULL
  ..$ : chr [1:5] "DTP1" "HepB_BD" "Pol3" "Hib3" "MCV1"...
```

(Note: DTP1 = First dose of Diphtheria/Pertussis/Tetanus vaccine (i.e., DTP); HepB_BD = Hepatitis B, Birth Dose (HepB); Pol3 = Polio third dose (Polio); Hib3 – Influenza third dose; MCV1 = Measles first dose (included in MMR, which is what California records))

districts – A sample of 700 California public school districts from the 2017 data collection, along with specific numbers and percentages for each district:

```
'data.frame':   700 obs. of  14 variables:
 $ DistrictName    : Name of the district
 $ WithDTP         : Percentage of students in the district with the DTP vaccine
 $ WithPolio       : Percentage of students in the district with the Polio vaccine
 $ WithMMR         : Percentage of students in the district with the MMR vaccine
 $ WithHepB        : Percentage of students in the district with Hepatitis B vaccine
 $ PctUpToDate     : Percentage of students with completely up-to-date vaccines
 $ DistrictComplete: Boolean showing whether or not district's reporting was complete
 $ PctBeliefExempt : Percentage of all enrolled students with belief exceptions
 $ PctMedicalExempt: Percentage of all enrolled students with medical exceptions
 $ PctChildPoverty : Percentage of children in district living below the poverty line
 $ PctFamilyPoverty: Percentage of families in district living below the poverty line
 $ PctFreeMeal     : Percentage of students in the district receiving free or
                     reduced cost meals (another indicator of poverty)
 $ Enrolled        : Total number of enrolled students in the district
 $ TotalSchools    : Total number of different schools in the district
```

As might be expected, the data are quite skewed: districts range from 1 to 582 schools enrolling from 10 to more than 50,000 students (NB. your sample may be slightly different). Further, while most districts have low rates of missing vaccinations, a handful are quite high. Be sure to note problems the data cause for the analysis and address any problems you can. Note that the data are about districts, not individual students, so be careful that you do not commit an ecological fallacy by stating conclusions about individuals.

In addition, you will find on Blackboard a CSV file, All Schools.csv, with data about 7,381 individual schools.

```
'data.frame' 7,381 obs. of 18 variables:
 $ SCHOOL CODE              : School ID number
 $ PUBLIC/ PRIVATE          : School status, "PUBLIC" or "PRIVATE" (note the space in the variable name
 $ Public School District ID: School district ID (only if public)
 $ PUBLIC SCHOOL DISTRICT   : School district name (only if public)
 $ CITY                     : City name
 $ COUNTY                   : Country name
 $ SCHOOL NAME              : School name
 $ ENROLLMENT               : Total number of enrolled students in the school
 $ UP_TO_DATE               : Number of students with completely up-to-date vaccines
 $ CONDITIONAL              : Number of students missing some vaccine without an exemption
 $ PME                      : Number of students with a medical exemption
 $ PBE_BETA                 : Number of students with a personal belief exemption
 $ DTP                      : Number of students in the district with the DTP vaccine
 $ POLIO                    : Number of students in the district with the Polio vaccine
 $ MMR                      : Number of students in the district with the MMR vaccine
 $ HEPB                     : Number of students in the district with Hepatitis B vaccine
 $ VARICELLA                : Number of students in the district with Varicella vaccine
 $ REPORTED                 : Whether the school reported vaccination data (Y or N)
```

Please document here your data exploration (univariate and bivariate) and any cleaning or transformations you find it necessary to apply, with concise explanations of what you did and why. Note any interesting patterns if you see them. At the end of the notebook, save the cleaned data to load at the start of the final exam notebook. Submit a knitted version of this notebook before the starting time of the final exam.

```r
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(tidyr)
library(readr)
setwd("/Users/katialeach/Downloads")
districts <- read.csv("districts.csv")
USAvaccinesTS <- read_csv("USAvaccinesTS.csv")
```

```
## Rows: 38 Columns: 6

## -- Column specification --------------------------------------------------------
## Delimiter: ","
## dbl (6): Year, DTP1, HepB_BD, Pol3, Hib3, MCV1
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
Public_School_Districts <- read_csv("Public School Districts.csv")
```

```
## New names:
## Rows: 846 Columns: 31
## -- Column specification
## -------------------------------------------------------- Delimiter: "," chr
## (18): School District Name, City, County, Up to date with vaccinations -... dbl
## (12): Enrolled students, Up to date with vaccinations, Conditional vacci... lgl
## (1): ...4
## i Use `spec()` to retrieve the full column specification for this data. i
## Specify the column types or set `show_col_types = FALSE` to quiet this message.
## * `` -> `...4`
```

```r
Public_SD <- Public_School_Districts
usVaccines <- read_csv("usVaccines.csv")
```

```
## Rows: 38 Columns: 5
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## dbl (5): DTP1, HepB_BD, Pol3, Hib3, MCV1
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
All_Schools <- read_csv("All Schools.csv")
```

```
## Warning: One or more parsing issues, call `problems()` on your data frame for details,
## e.g.:
##   dat <- vroom(...)
##   problems(dat)
```

```
## Rows: 7382 Columns: 18
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr  (7): PUBLIC/ PRIVATE, Public School District ID, PUBLIC SCHOOL DISTRICT...
## dbl (11): SCHOOL CODE, ENROLLMENT, UP_TO_DATE, CONDITIONAL, PME, PBE_BETA, D...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
library(BayesFactor)
```

```
## Loading required package: coda
## Loading required package: Matrix
##
## Attaching package: 'Matrix'
##
## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack
##
```

```
## ************
## Welcome to BayesFactor 0.9.12-4.7. If you have questions, please contact Richard Morey (richarddmorey
##
## Type BFManual() to open the manual.
## ************
```

Univariate exploration

```r
library(readr)
setwd("~/Downloads")
write.csv(districts, "~/Downloads/districts.csv", row.names = FALSE)
write.csv(usVaccines, "~/Downloads/usVaccines.csv", row.names = FALSE)
districts <- read_csv("districts.csv")
```

```
## Rows: 700 Columns: 14
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr  (1): DistrictName
## dbl (12): WithDTP, WithPolio, WithMMR, WithHepB, PctUpToDate, PctBeliefExemp...
## lgl  (1): DistrictComplete
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
summary(districts)
```

```
##  DistrictName          WithDTP         WithPolio      WithMMR
##  Length:700         Min.   : 23.00   Min.   : 23   Min.   : 23.00
##  Class :character   1st Qu.: 86.00   1st Qu.: 87   1st Qu.: 86.00
##  Mode  :character   Median : 93.00   Median : 94   Median : 94.00
##                     Mean   : 89.58   Mean   : 90   Mean   : 89.58
##                     3rd Qu.: 97.00   3rd Qu.: 97   3rd Qu.: 97.00
##                     Max.   :100.00   Max.   :100   Max.   :100.00
##
##     WithHepB       PctUpToDate     DistrictComplete PctBeliefExempt
##  Min.   : 23.00   Min.   : 23.00   Mode :logical    Min.   : 0.000
##  1st Qu.: 90.00   1st Qu.: 84.00   FALSE:39         1st Qu.: 1.000
##  Median : 96.00   Median : 92.00   TRUE :661        Median : 3.000
##  Mean   : 92.12   Mean   : 88.19                    Mean   : 5.724
##  3rd Qu.: 98.00   3rd Qu.: 96.00                    3rd Qu.: 7.000
##  Max.   :100.00   Max.   :190.00                    Max.   :77.000
##
##  PctMedicalExempt PctChildPoverty PctFamilyPoverty  PctFreeMeal
##  Min.   :0.0000   Min.   : 2.00   Min.   : 0.00    Min.   :  0.00
##  1st Qu.:0.0000   1st Qu.:13.00   1st Qu.: 5.00    1st Qu.: 33.00
##  Median :0.0000   Median :20.50   Median : 9.00    Median : 50.00
##  Mean   :0.1514   Mean   :22.25   Mean   :11.44    Mean   : 48.89
##  3rd Qu.:0.0000   3rd Qu.:29.00   3rd Qu.:15.00    3rd Qu.: 69.00
##  Max.   :8.0000   Max.   :72.00   Max.   :47.00    Max.   :100.00
##                                                    NA's   :246
```

```
##     Enrolled        TotalSchools
##  Min.   :   10.0   Min.   :  1.000
##  1st Qu.:   55.0   1st Qu.:  1.000
##  Median :  219.5   Median :  3.000
##  Mean   :  605.0   Mean   :  6.963
##  3rd Qu.:  655.5   3rd Qu.:  8.000
##  Max.   :54238.0   Max.   :582.000
##
```

```r
summary(Public_SD)
```

```
##  School District Name     City              County            ...4
##  Length:846           Length:846         Length:846         Mode:logical
##  Class :character     Class :character   Class :character   NA's:846
##  Mode  :character     Mode  :character   Mode  :character
##
##
##
##
##  Up to date with vaccinations - Percentage
##  Length:846
##  Class :character
##  Mode  :character
##
##
##
##
##  Conditional vaccination status - Percentage
##  Length:846
##  Class :character
##  Mode  :character
##
##
##
##
##  Permanent Medical Exemption - Percentage
##  Length:846
##  Class :character
##  Mode  :character
##
##
##
##
##  Personal Belief Exemption - Percentage
##  Length:846
##  Class :character
##  Mode  :character
##
##
##
##
##  Without Diphtheria, Tetanus, Pertussis - Percentage Without Polio - Percentage
##  Length:846                                          Length:846
##  Class :character                                    Class :character
```

```
##  Mode  :character                                     Mode  :character
##
##
##
##
##  Without Measles, Mumps, Rubella - Percentage Without Hepatitis B - Percentage
##  Length:846                                   Length:846
##  Class :character                             Class :character
##  Mode  :character                             Mode  :character
##
##
##
##
##  Without Varicella - Percentage Children 5-17 in poverty - Percentage
##  Length:846                     Length:846
##  Class :character               Class :character
##  Mode  :character               Mode  :character
##
##
##
##
##  Free/Reduced Price Meal Count K-12  - Percentage
##  Length:846
##  Class :character
##  Mode  :character
##
##
##
##
##  Families with income below poverty level - Percentage Enrolled students
##  Length:846                                            Min.   :   10.0
##  Class :character                                      1st Qu.:   50.5
##  Mode  :character                                      Median :  189.5
##                                                        Mean   :  592.7
##                                                        3rd Qu.:  652.0
##                                                        Max.   :54238.0
##                                                        NA's   :18
##  Up to date with vaccinations Conditional vaccination status
##  Min.   :    3.0              Min.   :   0.0
##  1st Qu.:   42.0              1st Qu.:   1.0
##  Median :  168.5              Median :   7.0
##  Mean   :  537.1              Mean   :  37.5
##  3rd Qu.:  599.8              3rd Qu.:  28.0
##  Max.   :46141.0              Max.   :7184.0
##  NA's   :18                   NA's   :18
##  Permanent Medical Exemption Personal Belief Exemption
##  Min.   : 0.000              Min.   :  0.00
##  1st Qu.: 0.000              1st Qu.:  1.00
##  Median : 0.000              Median :  6.00
##  Mean   : 1.046              Mean   : 17.11
##  3rd Qu.: 1.000              3rd Qu.: 19.00
##  Max.   :71.000              Max.   :842.00
##  NA's   :18                  NA's   :18
##  Diphtheria, Tetanus, Pertussis     Polio          Measles, Mumps, Rubella
```

```
##  Min.   :    3.00                Min.   :    3.00   Min.   :    3.0
##  1st Qu.:   42.75                1st Qu.:   43.75   1st Qu.:   42.0
##  Median :  170.50                Median :  170.50   Median :  169.5
##  Mean   :  548.46                Mean   :  551.39   Mean   :  549.9
##  3rd Qu.:  609.50                3rd Qu.:  610.75   3rd Qu.:  607.0
##  Max.   :48339.00                Max.   :48845.00   Max.   :48990.0
##  NA's   :18                      NA's   :18         NA's   :18
##   Hepatitis B        Varicella       Schools who reported (%) Total Schools
##  Min.   :    3.0   Min.   :    3.0   Length:846               Min.   :  1.000
##  1st Qu.:   44.0   1st Qu.:   44.0   Class :character         1st Qu.:  1.000
##  Median :  175.0   Median :  175.5   Mode  :character         Median :  3.000
##  Mean   :  563.4   Mean   :  566.5                            Mean   :  6.775
##  3rd Qu.:  611.8   3rd Qu.:  621.8                            3rd Qu.:  8.000
##  Max.   :51026.0   Max.   :51621.0                            Max.   :582.000
##  NA's   :18        NA's   :18
##  Coordinates        School District ID   Schools
##  Length:846         Min.   :600001     Length:846
##  Class :character   1st Qu.:609308     Class :character
##  Mode  :character   Median :622425     Mode  :character
##                     Mean   :624307
##                     3rd Qu.:633890
##                     Max.   :691137
##
```

```r
summary(usVaccines)
```

```
##       DTP1          HepB_BD           Pol3            Hib3
##  Min.   :81.00   Min.   :11.00   Min.   :24.00   Min.   :52.00
##  1st Qu.:89.75   1st Qu.:17.00   1st Qu.:90.00   1st Qu.:87.00
##  Median :97.00   Median :19.00   Median :93.00   Median :91.00
##  Mean   :94.05   Mean   :34.21   Mean   :87.16   Mean   :89.21
##  3rd Qu.:98.00   3rd Qu.:54.50   3rd Qu.:94.00   3rd Qu.:93.00
##  Max.   :99.00   Max.   :74.00   Max.   :97.00   Max.   :94.00
##       MCV1
##  Min.   :82.00
##  1st Qu.:90.00
##  Median :92.00
##  Mean   :91.24
##  3rd Qu.:92.00
##  Max.   :98.00
```

```r
summary(All_Schools)
```

```
##   SCHOOL CODE      PUBLIC/ PRIVATE    Public School District ID
##  Min.   : 100016   Length:7382        Length:7382
##  1st Qu.:6016406   Class :character   Class :character
##  Median :6043806   Mode  :character   Mode  :character
##  Mean   :5546463
##  3rd Qu.:6116404
##  Max.   :7105125
##  NA's   :1
##  PUBLIC SCHOOL DISTRICT     CITY               COUNTY
##  Length:7382            Length:7382        Length:7382
```

```
##  Class :character       Class :character   Class :character
##  Mode  :character       Mode  :character   Mode  :character
##
##
##
##
##  SCHOOL NAME          ENROLLMENT       UP_TO_DATE      CONDITIONAL
##  Length:7382        Min.   : 10.00   Min.   :  0.00   Min.   :  0.000
##  Class :character   1st Qu.: 41.00   1st Qu.: 34.00   1st Qu.:  0.000
##  Mode  :character   Median : 74.00   Median : 66.00   Median :  2.000
##                     Mean   : 75.99   Mean   : 68.56   Mean   :  4.931
##                     3rd Qu.:104.00   3rd Qu.: 95.00   3rd Qu.:  6.000
##                     Max.   :544.00   Max.   :350.00   Max.   :127.000
##                     NA's   :400      NA's   :400      NA's   :400
##       PME             PBE_BETA            DTP            POLIO
##  Min.   : 0.0000   Min.   :  0.000   Min.   :  0.0   Min.   :  0.00
##  1st Qu.: 0.0000   1st Qu.:  0.000   1st Qu.: 36.0   1st Qu.: 36.00
##  Median : 0.0000   Median :  1.000   Median : 68.0   Median : 68.00
##  Mean   : 0.1408   Mean   :  2.351   Mean   : 70.1   Mean   : 70.44
##  3rd Qu.: 0.0000   3rd Qu.:  3.000   3rd Qu.: 96.0   3rd Qu.: 97.00
##  Max.   :19.0000   Max.   :127.000   Max.   :395.0   Max.   :381.00
##  NA's   :400       NA's   :400       NA's   :400     NA's   :400
##       MMR              HEPB          VARICELLA        REPORTED
##  Min.   :  0.00   Min.   :  0.00   Min.   :  0.00   Length:7382
##  1st Qu.: 36.00   1st Qu.: 37.00   1st Qu.: 37.00   Class :character
##  Median : 68.00   Median : 70.00   Median : 71.00   Mode  :character
##  Mean   : 70.21   Mean   : 72.06   Mean   : 72.44
##  3rd Qu.: 97.00   3rd Qu.: 99.00   3rd Qu.: 99.00
##  Max.   :381.00   Max.   :387.00   Max.   :374.00
##  NA's   :400      NA's   :400      NA's   :400
```

Bivariate exploration

```r
modelDTP <- lm(districts$WithDTP ~ districts$PctChildPoverty + districts$PctFamilyPoverty)
modelPolio <- lm(districts$WithPolio ~ districts$PctChildPoverty + districts$PctFamilyPoverty)
modelMMR <- lm(districts$WithMMR ~ districts$PctChildPoverty + districts$PctFamilyPoverty)
modelHepB <- lm(districts$WithHepB ~ districts$PctChildPoverty + districts$PctFamilyPoverty)
bayesmodelDTP <- lmBF(WithDTP ~ PctChildPoverty + PctFamilyPoverty, data = districts)
```

```
## Warning: data coerced from tibble to data frame
```

```r
bayesmodelPolio <- lmBF(WithPolio ~ PctChildPoverty + PctFamilyPoverty, data = districts)
```

```
## Warning: data coerced from tibble to data frame
```

```r
bayesmodelMMR <- lmBF(WithMMR ~ PctChildPoverty + PctFamilyPoverty, data = districts)
```

```
## Warning: data coerced from tibble to data frame
```

```r
bayesmodelHepB <- lmBF(WithHepB ~ PctChildPoverty + PctFamilyPoverty, data = districts)
```

```
## Warning: data coerced from tibble to data frame
```

```r
summary(modelDTP)
```

```
##
## Call:
## lm(formula = districts$WithDTP ~ districts$PctChildPoverty +
##     districts$PctFamilyPoverty)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -64.418  -3.334   3.505   6.948  13.536
##
## Coefficients:
##                            Estimate Std. Error t value Pr(>|t|)
## (Intercept)                85.65972    0.87502  97.894  < 2e-16 ***
## districts$PctChildPoverty  -0.03120    0.06858  -0.455    0.649
## districts$PctFamilyPoverty  0.40321    0.10092   3.995 7.15e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10.78 on 697 degrees of freedom
## Multiple R-squared:  0.07081,    Adjusted R-squared:  0.06815
## F-statistic: 26.56 on 2 and 697 DF,  p-value: 7.656e-12
```

```r
summary(modelHepB)
```

```
##
## Call:
## lm(formula = districts$WithHepB ~ districts$PctChildPoverty +
##     districts$PctFamilyPoverty)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -64.755  -2.029   2.678   5.468  12.365
##
## Coefficients:
##                            Estimate Std. Error t value Pr(>|t|)
## (Intercept)                88.35658    0.77687 113.734  < 2e-16 ***
## districts$PctChildPoverty  -0.03006    0.06089  -0.494    0.622
## districts$PctFamilyPoverty  0.38790    0.08960   4.329 1.72e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9.573 on 697 degrees of freedom
## Multiple R-squared:  0.08211,    Adjusted R-squared:  0.07947
## F-statistic: 31.17 on 2 and 697 DF,  p-value: 1.078e-13
```

```r
summary(modelPolio)
```

```
##
## Call:
## lm(formula = districts$WithPolio ~ districts$PctChildPoverty +
##     districts$PctFamilyPoverty)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -62.888  -3.107   3.272   6.742  14.719
##
## Coefficients:
##                            Estimate Std. Error t value Pr(>|t|)
## (Intercept)                86.09534    0.86936  99.033  < 2e-16 ***
## districts$PctChildPoverty  -0.03395    0.06814  -0.498    0.618
## districts$PctFamilyPoverty  0.40756    0.10027   4.065 5.36e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10.71 on 697 degrees of freedom
## Multiple R-squared:  0.07203,    Adjusted R-squared:  0.06936
## F-statistic: 27.05 on 2 and 697 DF,  p-value: 4.856e-12
```

```r
summary(modelMMR)
```

```
##
## Call:
## lm(formula = districts$WithMMR ~ districts$PctChildPoverty +
##     districts$PctFamilyPoverty)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -64.353  -3.317   3.723   7.006  15.044
##
## Coefficients:
##                            Estimate Std. Error t value Pr(>|t|)
## (Intercept)                85.50422    0.89718  95.303  < 2e-16 ***
## districts$PctChildPoverty  -0.02283    0.07032  -0.325 0.745569
## districts$PctFamilyPoverty  0.40077    0.10348   3.873 0.000118 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11.06 on 697 degrees of freedom
## Multiple R-squared:  0.07036,    Adjusted R-squared:  0.06769
## F-statistic: 26.38 on 2 and 697 DF,  p-value: 9.08e-12
```

```r
summary(bayesmodelDTP)
```

```
## Bayes factor analysis
## --------------
## [1] PctChildPoverty + PctFamilyPoverty : 786662201 ±0%
##
```

```
## Against denominator:
##   Intercept only
## ---
## Bayes factor type: BFlinearModel, JZS
```

```r
summary(bayesmodelHepB)
```

```
## Bayes factor analysis
## --------------
## [1] PctChildPoverty + PctFamilyPoverty : 51496958456 ±0%
##
## Against denominator:
##   Intercept only
## ---
## Bayes factor type: BFlinearModel, JZS
```

```r
summary(bayesmodelPolio)
```

```
## Bayes factor analysis
## --------------
## [1] PctChildPoverty + PctFamilyPoverty : 1229355045 ±0%
##
## Against denominator:
##   Intercept only
## ---
## Bayes factor type: BFlinearModel, JZS
```

```r
summary(bayesmodelMMR)
```

```
## Bayes factor analysis
## --------------
## [1] PctChildPoverty + PctFamilyPoverty : 665484872 ±0%
##
## Against denominator:
##   Intercept only
## ---
## Bayes factor type: BFlinearModel, JZS
```

```r
t.test(districts$WithDTP, USAvaccinesTS$DTP1)
```

```
##
##  Welch Two Sample t-test
##
## data:  districts$WithDTP and USAvaccinesTS$DTP1
## t = -4.2981, df = 52.879, p-value = 7.428e-05
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -6.564141 -2.386837
## sample estimates:
## mean of x mean of y
##  89.57714  94.05263
```

```r
t.test(districts$WithMMR, USAvaccinesTS$MCV1)
```

```
##
##  Welch Two Sample t-test
##
## data:  districts$WithMMR and USAvaccinesTS$MCV1
## t = -2.057, df = 72.489, p-value = 0.04328
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -3.26231534 -0.05136887
## sample estimates:
## mean of x mean of y
##  89.58000  91.23684
```

```r
t.test(districts$WithPolio, USAvaccinesTS$Pol3)
```

```
##
##  Welch Two Sample t-test
##
## data:  districts$WithPolio and USAvaccinesTS$Pol3
## t = 1.1259, df = 39.13, p-value = 0.2671
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -2.264528  7.951595
## sample estimates:
## mean of x mean of y
##  90.00143  87.15789
```

```r
t.test(districts$WithHepB, USAvaccinesTS$HepB_BD)
```

```
##
##  Welch Two Sample t-test
##
## data:  districts$WithHepB and USAvaccinesTS$HepB_BD
## t = 15.756, df = 37.791, p-value < 2.2e-16
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  50.47132 65.35620
## sample estimates:
## mean of x mean of y
##  92.12429  34.21053
```

Data cleaning

```r
new_districts <- districts
new_Public <- Public_SD
new_all <- All_Schools
new_vaccines <- USAvaccinesTS
new_districts$PctFreeMeal <- ifelse(is.na(new_districts$PctFreeMeal),median(new_districts$PctFreeMeal,
```

```r
                                                                    na.rm = TRUE), new_districts$PctFreeMeal)
new_Public <- Public_SD[, -4]
new_Public <- new_Public %>% mutate(across(where(is.numeric), ~replace_na(., mean(., na.rm=TRUE))))
new_all <- new_all %>% mutate(across(where(is.numeric), ~replace_na(., mean(., na.rm=TRUE))))
new_vaccines <- new_vaccines %>% mutate(across(where(is.numeric), ~replace_na(., mean(., na.rm=TRUE))))
sum(is.na(new_districts))
```

```
## [1] 0
```

```r
colSums(is.na(new_Public))
```

```
##                                  School District Name
##                                                     0
##                                                  City
##                                                     0
##                                                County
##                                                     0
##             Up to date with vaccinations - Percentage
##                                                    18
##           Conditional vaccination status - Percentage
##                                                    18
##              Permanent Medical Exemption - Percentage
##                                                    18
##                Personal Belief Exemption - Percentage
##                                                    18
##   Without Diphtheria, Tetanus, Pertussis - Percentage
##                                                    18
##                            Without Polio - Percentage
##                                                    18
##          Without Measles, Mumps, Rubella - Percentage
##                                                    18
##                      Without Hepatitis B - Percentage
##                                                    18
##                        Without Varicella - Percentage
##                                                    18
##                 Children 5-17 in poverty - Percentage
##                                                    56
##      Free/Reduced Price Meal Count K-12  - Percentage
##                                                     1
## Families with income below poverty level - Percentage
##                                                    59
##                                     Enrolled students
##                                                     0
##                          Up to date with vaccinations
##                                                     0
##                        Conditional vaccination status
##                                                     0
##                           Permanent Medical Exemption
##                                                     0
##                             Personal Belief Exemption
##                                                     0
##                        Diphtheria, Tetanus, Pertussis
##                                                     0
```

```
##                                          Polio
##                                              0
##                            Measles, Mumps, Rubella
##                                              0
##                                       Hepatitis B
##                                              0
##                                         Varicella
##                                              0
##                          Schools who reported (%)
##                                              0
##                                     Total Schools
##                                              0
##                                       Coordinates
##                                              0
##                                 School District ID
##                                              0
##                                           Schools
##                                              0
```

```
colSums(is.na(new_all))
```

```
##               SCHOOL CODE           PUBLIC/ PRIVATE Public School District ID
##                         0                         1                         1
##    PUBLIC SCHOOL DISTRICT                      CITY                    COUNTY
##                         1                         1                         1
##               SCHOOL NAME                ENROLLMENT                UP_TO_DATE
##                         1                         0                         0
##               CONDITIONAL                       PME                  PBE_BETA
##                         0                         0                         0
##                       DTP                     POLIO                       MMR
##                         0                         0                         0
##                      HEPB                 VARICELLA                  REPORTED
##                         0                         0                         1
```

```
colSums(is.na(new_vaccines))
```

```
##    Year    DTP1 HepB_BD    Pol3    Hib3    MCV1
##       0       0       0       0       0       0
```

```
write.csv(new_Public, "~/Downloads/new_Public.csv", row.names = FALSE)
write.csv(new_districts, "~/Downloads/new_districts.csv", row.names = FALSE)
write.csv(new_all, "~/Downloads/new_all.csv", row.names = FALSE)
write.csv(new_vaccines, "~/Downloads/new_vaccines.csv", row.names = FALSE)
```

Transformations

I replaced NA with the column means in the columns that needed it. I saved the data as csv and change character to numeric and then repplaced NAs. I wasn't sure what to do for the analysis so I did only some analysis.

Notes for analysis

Polio vaccination in the us vs the california districts in not statistically significant. It could be due that polio is virtually iradicated in the US, thus the vaccince isn't as relevant. Low HepB rates could be due to lack of education on the disease, poor public sentiment, poor access to health care, etc.

The adjusted R2 in the linear models is extremely low.